

Statistics II: Introduction to Inference

Module 7: Interval Estimation

Sometimes providing a point estimate, or testing a hypothesis is not the ideal method of inference. Rather, one may be interested in an interval (or a set), which *efficiently* captures the underlying parameter. For example, in a diagnostic test for a particular disease, one requires an interval (range) of the possible test results, with efficiently detects the occurrence or non-occurrence of a particular disease. In other words we require a random set which captures the underlying parameter with high probability (rather than a random point which is close to the underlying parameter in appropriate sense). This type of estimates are called interval estimates.

Definition 1 (Interval Estimator). Let $X_1, \dots, X_n \sim F_\theta$. An interval estimator for a function of the parameter $g(\theta)$ is a pair of functions of sample $L(\mathbf{X}) = L(X_1, \dots, X_n)$, $U(\mathbf{X}) = U(X_1, \dots, X_n)$ that satisfy $L(\mathbf{x}) \leq U(\mathbf{x})$ for each $\mathbf{x} \in \mathbb{R}^n$. If realization of $\mathbf{X}$ is $\mathbf{x}$, then we refer to $[L(\mathbf{x}), U(\mathbf{x})]$ as an *interval estimate* of $g(\theta)$.

Note: For some particular examples, $L(\mathbf{x})$ can be $-\infty$, or $U(\mathbf{x})$ can be ∞ . In such cases we obtain a one-sided interval estimate. Further, instead of closed interval, one may obtain an open, or a semi-closed interval estimate as well.

Definition 2 (Confidence Coefficient). The confidence coefficient of an interval estimator $[L(\mathbf{X}), U(\mathbf{X})]$ of a parameter $g(\theta)$, usually denoted by $(1 - \alpha)$, is highest probability with which the random interval captures the true parameter function $g(\theta)$, for any $\theta \in \Theta$. Notationally, the confidence coefficient is a number $\beta = (1 - \alpha) \in (0, 1)$ satisfying

$$\inf_{\theta} P_{\theta \in \Theta} (L(\mathbf{X}) \leq g(\theta) \leq U(\mathbf{X})) = (1 - \alpha), \text{ so that, } P_{\theta \in \Theta} (L(\mathbf{X}) \leq g(\theta) \leq U(\mathbf{X})) \geq (1 - \alpha), \text{ for all } \theta \in \Theta.$$

Interval estimators together with confidence coefficient are called **confidence intervals**. We write, $[L(\mathbf{X}), U(\mathbf{X})]$ is a $100(1 - \alpha)\%$ confidence interval of $g(\theta)$.

Note: We can generalize the idea of confidence intervals to *confidence sets*. A random set $S(\mathbf{X})$ is said to be a confidence set for a function of parameter vector $\mathbf{g}(\theta) \in \Theta \subseteq \mathbb{R}^k$, with confidence coefficient $(1 - \alpha)$, if $\inf_{\theta \in \Theta} P_{\theta} (S(\mathbf{X}) \ni \mathbf{g}(\theta)) = (1 - \alpha)$. A confidence interval can be interpreted as a special type of confidence set, where $S(\mathbf{X})$ is an interval.

Interpretation of Confidence Intervals: A confidence interval $[L(\mathbf{X}), U(\mathbf{X})]$ of $g(\theta)$ with confidence coefficient $(1 - \alpha)$ can be interpreted as follows:

If repeated random samples, that is, repeated realizations of $\mathbf{X}$, are taken for a large (theoretically, infinite) number of times, then in about $(1 - \alpha)100\%$ cases, the realization of the confidence set, $[L(\mathbf{x}), U(\mathbf{x})]$, would contain the true parameter $g(\theta)$.

Example 1. Let $X_1, \dots, X_n$ be a random sample from $\text{Normal}(\mu, \sigma^2)$ distribution. Consider the interval estimate $[\bar{X}_n - c, \bar{X}_n + c]$ of μ for some constant $c \geq 0$. Find c such that the confidence coefficient is $(1 - \alpha)$.

When σ^2 is known, then it can be seen that $c = \sigma \tau_{\alpha/2} / \sqrt{n}$ where $\tau_{\alpha/2}$ is the upper $\alpha/2$ -th point of the standard normal distribution.

Remark 1. One could also choose an interval estimate of the type $[\bar{X}_n - c_1, \bar{X}_n + c_2]$ where $c_1, c_2 \geq 0$. Then any c_1, c_2 , satisfying $\Phi(c_1 \sqrt{n}/\sigma) - \Phi(-c_2 \sqrt{n}/\sigma) = (1 - \alpha)$ would provide a valid confidence interval with confidence coefficient $(1 - \alpha)$.

Remark 2. When σ^2 is unknown then one can use the fact that $\sqrt{n}(\bar{X}_n - \mu)/S_n^* \sim t_{(n-1)}$ to obtain find c_1, c_2 , where $S_n^{*2} = (n - 1)^{-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$.

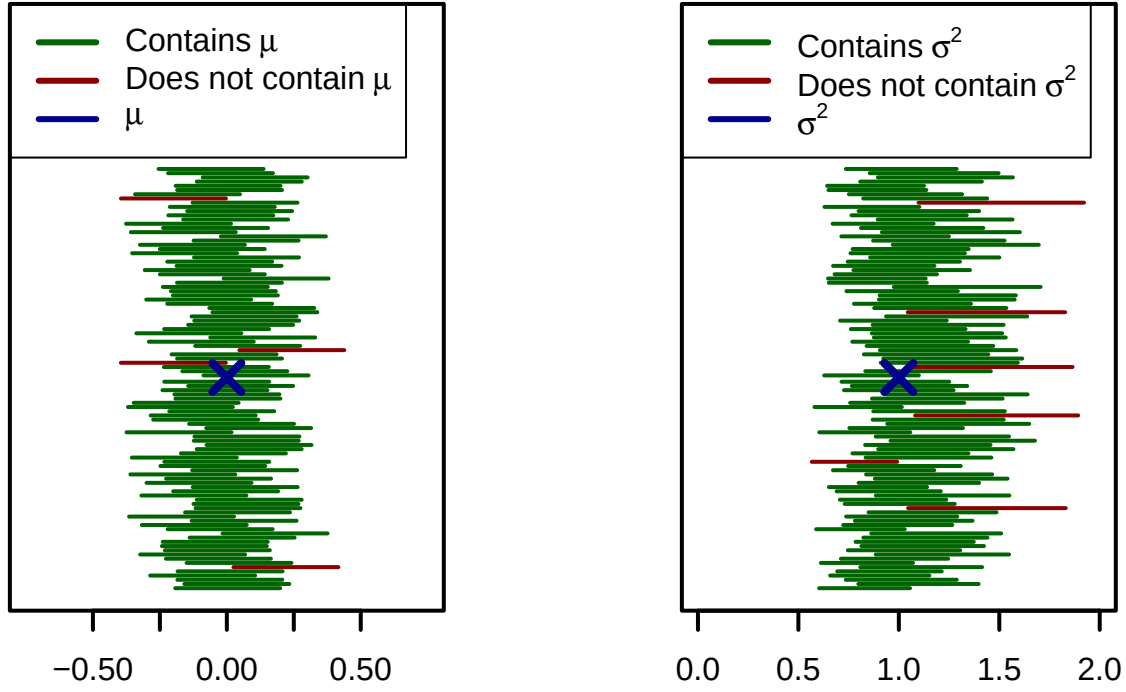


Figure 1: Figure contains 100 realizations of the 95% confidence intervals from normal mean μ (left) and variance σ^2 (right) as obtained in Examples 1 and 2. The realizations of the interval estimates which captures (does not capture) the true parameter are indicated in green (red), while the true parameters are indicated in blue. Observe that, in about 95% cases, the realization of the interval contains the true parameter.

1 Method of Finding Confidence Interval

1.1 Method of Pivots

Definition 3 (Pivot). Let $\mathbf{X} \sim f_{\mathbf{X}}(\cdot; \theta)$. A random variable $T(\mathbf{X}; \theta)$ is called a pivot if the distribution of $T(\mathbf{X}; \theta)$ does not depend on θ .

Example 2. Let $X_1, \dots, X_n$ be a random sample from a location family with location parameter θ , i.e., $X_i = W_i + \theta$ where W_i , $i = 1, \dots, n$ are iid from a distribution free of θ . Then any function of $X_i - \theta$; $i = 1, \dots, n$ is a pivot.

Example 3. Let $X_1, \dots, X_n$ be a random sample from a scale family with scale parameter θ , i.e., $X_i = \theta W_i$ where W_i , $i = 1, \dots, n$ are iid from a distribution free of θ . Then any function of X_i/θ ; $i = 1, \dots, n$ is a pivot.

Example 4. Let $X_1, \dots, X_n$ be a random sample from a location-scale family with location parameter θ_1 and scale parameter θ_2 , i.e., $X_i = \theta_1 + \theta_2 W_i$ where W_i , $i = 1, \dots, n$ are iid from a distribution free of (θ_1, θ_2) . Then any function of $(X_i - \theta_1)/\theta_2$; $i = 1, \dots, n$ is a pivot.

Example 5. If X is a continuous random variable, then the distribution function of X , $U_Y(\theta) = F_X(Y; \theta) = P_\theta(X \leq Y)$ has a **Uniform**(0,1) distribution. When n i.i.d. samples $X_1, \dots, X_n$ are available, then one may take $T(\mathbf{X}; \theta) = -\sum_{i=1}^n \log F_X(X_i; \theta)$ as a pivot. It can be shown that $T(\mathbf{X}; \theta)$ distributed as a **Gamma**($n, 1$) distribution.

A pivot may yield a confidence interval when supported with some additional features. The following theorem provides a set of sufficient conditions for a pivot to yield a confidence interval.

Theorem 1. Let $T(\mathbf{X}; \theta)$ be a pivot, such that

- i. for each $g(\theta)$, $T(\mathbf{X}; \theta)$ is a statistic (i.e., if $g(\theta) = g_0$ is given, then $T(\mathbf{X}; \theta)$ becomes a function of $X_1, \dots, X_n$ and g_0),
- ii. as a function of $g(\theta)$, $T(\mathbf{x}; \theta)$ is strictly monotone at each $\mathbf{x} \in \mathbb{R}^n$.
- iii. Let $\Lambda \in \mathbb{R}$ be the range of $T(\mathbf{X}; \theta)$, and for each $\lambda \in \Lambda$ and $\mathbf{x} \in \mathbb{R}$ the equation $\lambda = T(\mathbf{x}; \theta)$ is solvable with respect to $g(\theta)$.

Then one can construct a confidence interval for θ at any level.

Procedure of obtaining confidence interval from a pivot. Suppose we are interested in finding a confidence interval for the parameter-function of interest $g(\theta)$ based on i.i.d. samples $X_1, \dots, X_n \sim F_\theta$.

1. First find a pivot $T(\mathbf{X}, \theta)$ such that $T(\mathbf{X}, \theta)$ is a function of $g(\theta)$ and a good point estimator of $g(\theta)$.
2. Find constants a, b such that $P(a \leq T(\mathbf{X}, \theta) \leq b) = (1 - \alpha)$. Note that, as the distribution of $T(\mathbf{X}, \theta)$ is completely known, the constants (a, b) are also known (does not depend on θ).
3. Next, solve the equations $T(\mathbf{X}, \theta) = a$ and $T(\mathbf{X}, \theta) = b$ with respect to $g(\theta)$. Let $S_n(a)$ and $S_n(b)$ be the solutions, i.e., $T(\mathbf{X}, \theta) = a \Leftrightarrow g(\theta) = S_n(a)$ and $T(\mathbf{X}, \theta) = b \Leftrightarrow g(\theta) = S_n(b)$.
Note that, $(S_n(a), S_n(b))$ are functions of $\mathbf{X}$ and (a, b) only. Thus, they are pair of statistics.
4. Let $T(\mathbf{X}, \theta)$ be strictly monotonically increasing function of $g(\theta)$. Then

$$\{a \leq T(\mathbf{X}, \theta) \leq b\} \iff \{S_n(a) \leq g(\theta) \leq S_n(b)\},$$

so that $P_\theta(S_n(a) \leq g(\theta) \leq S_n(b)) = (1 - \alpha)$ for all $\theta \in \Theta$. Thus, $[S_n(a), S_n(b)]$ is a confidence interval for $g(\theta)$ with confidence coefficient $(1 - \alpha)$.

Further, let $T(\mathbf{X}, \theta)$ be strictly monotonically decreasing function of $g(\theta)$. Then

$$\{a \leq T(\mathbf{X}, \theta) \leq b\} \iff \{S_n(b) \leq g(\theta) \leq S_n(a)\},$$

so that $P_\theta(S_n(b) \leq g(\theta) \leq S_n(a)) = (1 - \alpha)$ for all $\theta \in \Theta$. Thus, $[S_n(b), S_n(a)]$ is a confidence interval with confidence coefficient $(1 - \alpha)$.

Remark 3. A sufficient condition for the equation $\lambda = T(\mathbf{x}; \theta)$ to be solvable is T is continuous and strictly monotone w.r.t. θ .

Remark 4. The monotonicity assumption in the above theorem ensures that the confidence set obtained from the pivot T is of interval type. In case all the other assumptions in Theorem 1 are satisfied, except the monotonicity assumption, then one would still obtain a confidence set, but it may not be of interval type.

Example 6 Let $X_1, \dots, X_n$ be a random sample from $\text{Normal}(\mu, \sigma^2)$ distribution. We need to find a confidence interval for σ^2 . As $S_n^{*2} = (n-1)^{-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$ is the UMVUE of σ^2 , we may find a pivot based on S_n^{*2} .

Recall the result that $T_n(\sigma^2) = (n-1)\sigma^{-2}S_n^{*2} \sim \chi_{(n-1)}^2$, which can be an appropriate pivot, as i. $T_n(\sigma^2)$ is a function of S_n^{*2} and σ^2 only; ii. $T_n(\sigma^2)$ is strictly decreasing function of σ^2 for each realization of S_n^{*2} , iii. $T_n(\sigma^2) = \lambda$ is solvable with respect to σ^2 for each $\lambda > 0$.

Next we find (a, b) such that

$$P(a \leq T_n(\sigma^2) \leq b \mid T_n(\sigma^2) \sim \chi_{(n-1)}^2) = (1 - \alpha).$$

In particular, we may choose $a = \chi_{1-\alpha/2; n-1}^2$ and $b = \chi_{\alpha/2; n-1}^2$, where $\chi_{\alpha; n-1}^2$ is the upper α point of χ_{n-1}^2 distribution.

Next, we solve the equations $T_n(\sigma^2) = a$ and $T_n(\sigma^2) = b$ for σ^2 , and obtain $S_n(a) = (n-1)S_n^{*2}/a = (n-1)S_n^{*2}/\chi_{1-\alpha/2; n-1}^2$ and $S_n(b) = (n-1)S_n^{*2}/b = (n-1)S_n^{*2}/\chi_{\alpha/2; n-1}^2$.

As $T_n(\sigma^2)$ is a strictly decreasing function of σ^2 , a $100(1 - \alpha)\%$ confidence interval for σ^2 is $[S_n(b), S_n(a)] = [(n-1)S_n^{*2}/\chi_{\alpha/2; n-1}^2, (n-1)S_n^{*2}/\chi_{1-\alpha/2; n-1}^2]$